

Tairan Song

Email: songtail@msu.edu / tairan_song@outlook.com

Tel: (+1) 734-819-0577 Website: tairan-song.github.io

Education

Michigan State University , East Lansing, MI, USA	Aug 2023 – Present
Ph.D. Student in Computational Mathematics, Science and Engineering	
M.S. in Computational Mathematics, Science and Engineering; GPA: 3.8 / 4.0	Aug 2023 – Aug 2026
Advisor: Dr. Adam Alessio	
King's College London , London, Greater London, United Kingdom	Sep 2021 – Sep 2022
M.Sc. in Applied Statistical Modelling and Health Informatics; Graduated with Distinction	
Advisors: Dr. Ioannis Bakolis and Dr. Amy Ronaldson	
University of Liverpool , Liverpool, Merseyside, United Kingdom	Sep 2018 – Jun 2021
B.Sc. in Mathematics and Statistics; Upper Second-Class Honours	
Advisor: Dr. Jinlai Li	

Publications

Zhang, T., Jia, H., **Song, T.**, Lv, L., Gulhan, D. C., Wang, H., ... & Shen, N. (2023). De novo identification of expressed cancer somatic mutations from single-cell RNA sequencing data. *Genome Medicine*, 15(1), 115. doi: 10.1186/s13073-023-01269-1

Conference Presentations & Abstracts

- **Song, T.**, et al. (2026). *Core-to-Margin Radiomic Profiling for Predicting Kidney Cancer Grade from Variable-Phase CT*. Accepted scientific poster presentation at the Radiological Society of North America (RSNA) 2026 Annual Meeting, Chicago, IL, November 29–December 3, 2026.
- Muterspaugh, R., Munavar Ali, M. A., **Song, T.**, et al. (2026). *Deep Learning-Based Detection of High-Risk Renal Masses from All-Cause Abdominal CT*. Abstract presented at *Corewell Health West Research Day*, Grand Rapids, MI, May 8, 2026. Abstract 1968.
- **Song, T.** (2024). *Enhancing Transcription Factor Binding Site Prediction with Epigenetic Data: A BERT-Based Machine Learning Innovation*. Poster presented at the 20th Annual Meeting of the MidSouth Computational Biology and Bioinformatics Society (MCBIOS 2024), Emory University, Atlanta, GA, March 22–24, 2024.

Research Experience

Spatial Graph Modeling of Renal Tumor Heterogeneity from Abdominal CT

Advisor: Dr. Adam Alessio Michigan State University Jan 2026 – Present

- Developing graph-based representations of renal masses from abdominal CT to model spatial heterogeneity across tumor core, boundary, and peri-tumoral regions for interpretable renal tumor risk stratification.

Kidney Cancer Grade from Highly Variable Contrast Timing CT

Advisor: Dr. Adam Alessio Michigan State University Jan 2025 – Dec 2025

- Developed a radiomics-based modeling framework for renal mass grading from heterogeneous abdominal CT, explicitly addressing variability introduced by inconsistent contrast timing across real-world imaging protocols, and implemented an end-to-end pipeline with automatic multi-organ segmentation (nnU-Net for renal regions and TotalSegmentator for vascular structures), high-dimensional radiomics extraction, and supervised feature reduction.
- Proposed a reference-region-aware feature representation that augments tumor radiomics with quantitative descriptors from surrounding anatomical regions (normal kidney, aorta, inferior vena cava), improving robustness to contrast-timing variation without explicit phase normalization.
- Systematically evaluated multiple feature reduction strategies and classifiers under cross-validation, demonstrating consistent performance gains over tumor-only models for both multi-class and binary grading tasks.

De Novo Identification of Expressed Cancer Somatic Mutations from scRNA Sequencing Data

Advisor: Dr. Ning Shen Zhejiang University Sep 2022 – Mar 2023

- Contributed to the development and evaluation of the RESA (Recurrently Expressed SNV Analysis) computational framework for high-precision de novo identification of expressed somatic mutations from scRNA-seq data, which consistently outperforms existing methods across multiple benchmark datasets.
- Implemented and validated the joint logistic regression component (RESA-jLR) within the RESA pipeline to expand the pool of detectable somatic variants and improve mutation calling performance.
- Performed extensive data preprocessing, mutation calling workflows, and empirical performance evaluation, and generated publication-ready figures supporting sensitivity and precision benchmarking against other methods reported in the study.

Teaching Experience

Teaching Assistant	Michigan State University
• CMSE 202: Computational Modeling and Data Analysis II	Jan 2025 – May 2025

Research and Professional Experience

Zhejiang University	Hangzhou, China
Research Assistant	Sep 2022 – Mar 2023
China Galaxy Securities Co., Ltd.	Shanghai, China
Financial Data Analytics Intern	Jul 2021 – Sep 2021
China Post Group Co., Ltd., Hefei Branch	Hefei, China
Operations Data Analytics Intern	Jun 2018 – Sep 2018

Awards and Honors

NSF NRT-IMPACTS Training Support	Michigan State University	Fall 2023
Received one-semester graduate training support through the NSF Research Traineeship-funded IMPACTS Program (NSF DGE-1828149).		

Skills

Programming: Python, R, SQL
ML & AI: PyTorch, scikit-learn; deep learning for medical imaging (CNN-based architectures, e.g., U-Net); NLP and Transformer-based models
MLOps & Systems: SLURM, Git, High Performance Computing
Research: Medical Imaging Analysis, Biological sequence Analysis, Statistical Analysis, Interdisciplinary Research